

NIRANJAN VSKS

Senior Agentic AI Engineer | Forward Deployed Engineering

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Summary

Senior Agentic AI Engineer with **7 years** of expertise. I build **production agentic systems** that run inside the **customer's own cloud**, under their identity, their constraints and their data boundaries. I own the layer that keeps them **reliable and efficient at scale**: Finetuning, evaluation, latency and token cost. Blank page to production. **Core backend and systems engineering underneath**. The engineer who **architects, builds and deploys** from scratch.

Experience

Senior Engineer - GenAI & Agentic AI, Coforge

03/2024 – Present | Hyderabad

- **Architected and shipped** an **AI-infused Agentic Quality Engineering Platform end to end**, taking it from feasibility framing through go-live to production with **17 enterprise QA teams at peak adoption**; deployed it as **containerised services on Linux VMs** inside customer-managed **AWS, Azure and GCP** tenants, with **architecture**, compute **sizing** and identity through **Azure AD, IAM and SSO** adapted per cloud, all from one reference architecture rather than customer-specific forks.
- Cut production hallucination from **~15% to under 5%** with **GraphRAG on Neo4j** built over **live Jira and Azure DevOps data: ingestion, entity normalization**, query transformation, **re-ranking and context compression**, so generation is scoped against linked stories, epics and **acceptance criteria** rather than flat text, **with retrieval and response quality tracked as standing metrics** rather than a **one-time benchmark**.
- **Fine-tuned LLaMA and Qwen models with LoRA and aligned them with RLHF**. For a finance-sector client, adapted their **in-house LLM** with **LoRA** so their **own model** generated test cases directly inside the platform, under the **data-residency constraints** that ruled out hosted providers.
- **Owned the production ML lifecycle as one problem: token and cost engineering** (context caching, prompt-payload reduction, tiered model routing) underneath a live evaluation and FinOps layer with output quality KPIs I defined from scratch (Acceptance Criteria Coverage, Test Design Coverage, RAGAS), **model and retrieval quality monitored continuously against those thresholds**, plus per-tenant cost telemetry with per-request attribution that stakeholders read directly.
- Built a **custom MCP server** exposing browser control as typed tools, so agents turn natural-language intent into production-grade **Playwright and Cypress** scripts executed inside the customer's own CI; eliminated **85 to 90% of manual QA effort** with agentic test generation, execution and **CI-integrated automation**, and cut test authoring time **~75%**.
- **Lead developer and architect** on the platform, **directing the engineering team** from architecture through delivery and owning the technical decisions end to end across **three cloud rollouts**; drove the **cross-team migration** to a **single reference architecture** across three clouds without formal authority over the teams involved, and set the **code review, RFC and release standards** the practice now works to.
- The practice's **AI technical authority: L1 technical interviewer** for **AI/ML, GenAI & Agentic AI** hiring across **50+ interviews**; setting the technical bar, and **mentoring engineers** on agent design, retrieval architecture and **evaluation** practice.

Data Scientist - GenAI, ML & Conversational AI, Hewlett Packard Enterprise

11/2022 – 12/2023 | Bangalore

- Architected the migration of an enterprise conversational system from **rule-based FAQ to LLM-assisted RAG** over ~1,500-1,700 internal documents, cutting repeat user escalations **~40%**.
- Designed the **retrieval and ranking pipeline** end to end: chunking with **overlap calibration**, **hybrid search** combining **semantic and keyword** signals, **BM25** and metadata filtering tuned for enterprise-jargon queries. Added evaluation loops that measured **retrieval precision** against a curated held-out **evaluation** set, separating **retrieval** error from **generation** error, and improved response relevance **30-40% in pilot**.
- Built **Go plugin services (Gin, Kratos)** that extended the **enterprise RAG agent** with new capabilities, communicating with the agent over **RPC and gRPC** through **typed service contracts**.

Associate Software Engineer - Machine Learning & AI, Mphasis

07/2019 – 10/2022 | Bangalore

- Built **fraud-detection and anomaly-detection models** for financial services clients using tree based ensembles (**XGBoost, LightGBM, Random Forest**) with **SQL-based** feature extraction over **transactional data**, improving **precision ~15%** over legacy rule-based systems and cutting manual review volume.
- **Diagnosed and corrected the class imbalance** that makes fraud data misleading on a naive split, where a model scores 99% accuracy and catches nothing. **Chose precision as the operating metric** over accuracy and recall because the cost asymmetry sat on false positives, and engineered the behavioural and transactional features the models ran on from raw client data in **pandas**, working across **scikit-learn, PyTorch and TensorFlow**. Owned these models in production after launch, **retraining against drift** as transaction patterns shifted and **validating that gains held on held-out data** rather than on the training split.

Education

Product Management in Generative & Agentic AI, BITS School of Management (BITSOM)

07/2025 – 03/2026 | India

PG Diploma in Machine Learning & Artificial Intelligence, IIIT Bangalore

06/2020 – 03/2022 | Bangalore

Bachelor of Technology in Computer Science, Bharat Institute of Engineering and Technology

08/2015 – 06/2019 | Hyderabad

Skills

Software Engineering & Cloud: Python | Go (Gin, Kratos) | TypeScript | FastAPI | Node.js | PostgreSQL | MongoDB | Redis | Event-Driven and Async Architecture | Kafka | Microservice Architecture | Distributed Systems | Service Design and API Boundaries | Docker | Kubernetes | GitHub Actions | CI/CD | Terraform | AWS | GCP | Azure | Observability (Tracing, Metrics, Structured Logging) | FinOps and Cost Attribution | Code Review and Release Practice

Generative and Agentic AI: LLM Fine-Tuning (LoRA, RLHF) | LLMs (Azure OpenAI, Claude, Gemini, LLaMA, Qwen) | Multi-Agent Orchestration | Agentic Workflows | LangGraph | LangChain | LlamaIndex | MCP Servers | Tool and Function Calling | Structured Outputs | Prompt and Context Engineering | Model Gateway (Multi-Provider Routing, Fallback, Cost Attribution) | LLM Observability | Evaluation Design and Rubrics | RAGAS | Guardrails, AI Safety and Governance

Retrieval & Data Architecture: Knowledge Graphs | GraphRAG | Neo4j | AWS Neptune | Entity Resolution | Hybrid Search | Semantic Querying | Reranking | Vector Databases | Agent Memory Architecture | Agentic RAG Pipelines | pgvector | Citations & Grounding

Enterprise and Customer Facing: Forward Deployed Engineering | Solutions Architecture | Customer Discovery | Stakeholder Alignment | Microsoft Dynamics 365 Integration | Financial Systems Integration | Approval Workflows & Policy Controls | Audit Trails | OAuth2, JWT & Enterprise Authentication | Deployment into Customer Tenants (AWS, Azure, GCP) | Multi-Tenant Architecture | Identity & SSO Architecture | RBAC Design | Enterprise API Integration | Customer-Facing Technical Communication | Regulated Environment Deployment (RBAC, audit, data residency)

Projects

- Agentic Codebase Intelligence System**, *Multi-Agent Platform for Enterprise & Architectural Level Codebase Auditing* 06/2026 – Present
- Architected and shipped a **multi-agent codebase-intelligence platform** end to end that ingests **polyglot repositories** and builds a **persistent structural model** of the system (entities, **call paths**, **cross-file dependencies using ReAct**) that agents reason over, rather than retrieving flat file text. Cut manual review and audit effort **~50%**.
 - Built the **verification harness** the agents run inside and the **closed feedback loop** around it: deterministic checks, a **taste gate** that fails output on quality rather than correctness, and **one guard per defect class**, so every defect found in production becomes a permanent automated check and the system's reliability compounds instead of regressing.
 - Built an **audit orchestrator** coordinating **6 specialized agents** (retrieval, reasoning, verification) with **MCP tool integration**, compressing time-to-findings from **~2 days of manual review to under 20 minutes** and auto-filing prioritized issues into **Jira**.
- Loop Copilot: AI CRM Copilot for Enterprise Sales**, *Architect · Engineer · Sole Owner | Production, Fortune 500 pilot*
- Built and shipped a **production AI CRM copilot** for enterprise sales reps as **sole architect and engineer**; took it from **customer discovery through V2** with an active beta user inside a **Fortune 500 sales org**, cutting CRM logging time from **4-6 min to ~45 sec (~85%)**.
 - Shipped a **voice agent pipeline** that listens to sales-client calls (**STT**), turns them into structured **PRD and onboarding documents**, and **auto-detects** any newly mentioned accounts. It resolves them against the account repository using **unique ID**, then logs them straight into **D365** or the connected **CRM**, so reps never have to manually key in account details after a call.
 - Architected the full stack (**React 19, async FastAPI, MongoDB, Azure OpenAI, Entra ID via MSAL, Microsoft Graph**, Telegram Bot API, **CI/CD**) and chose **structured event sourcing** over vector RAG for memory, giving an **append-only auditable record** while removing hallucinated recall and keeping **cross-session context**; shipped **10+ feature areas** with a **multi-CRM expansion shell** (**Salesforce, HubSpot, Zoho**) architected and ready.
- Autonomous Wealth Operating System**, *Multi-Asset Agentic Application | Architect. Engineer. Owner* Ongoing
- Architected and built an **autonomous multi-asset wealth platform** (stocks, mutual funds, ETFs, bonds, NPS) around a **21-agent analyst council**: isolated agent opinions, a **devil's advocate stage**, aggregation weighted by each agent's **historical accuracy**, **ReAct** and **self-reflection** and a **code-enforced veto gate** so **no money moves on LLM output alone**.
 - Engineered a **five-ring risk system** (position, portfolio, drawdown, budget, and connection-level **broker locks**) plus a **frozen-threshold evolution gauntlet**: any strategy change must beat the incumbent across **fixed statistical gates** before promotion, which rules out **silent strategy drift**.
 - Delivered it as a **multi-surface platform**: a shared **framework-free core** behind web and mobile clients, a **Telegram copilot** for conversational instructions and live trade, investment and report updates, **user-selectable LLM models per agent role**, and wallets integrated with **Stripe** and **banking APIs**.
- Voice-First Financial AI Copilot**, *Capstone, BITSOM | Architected, Built & Piloted*
- Owned architecture and delivery of a **voice-first financial learning and advisory copilot** for India's gig workforce, from customer research through pilot: a **~30-screen MVP** with **10-language architecture**, shipped on both **web and mobile**.
 - Designed a **multi-layer compliance architecture** with **SEBI advisory principles** and **DPDP requirements** as **first-class design constraints**: **RAG over a regulatory-safe knowledge base**, **system-prompt guardrails**, server-side governance, and **consent-first data handling**.
 - Engineered a **central memory layer with RAG** so user context, goals, and prior conversations persist **across sessions and across web and mobile** from a **single source of truth**.

Certifications

Claude Certified Architect Anthropic : In-Progress	Agile Practices Certificate SAFe
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